Introduction to Python

Python is a high-level, interpreted programming language known for its simplicity and readability.

1 Java vs Python

Both programs solve the same problem:
Find the sum of even numbers in a list.

Java

```java
public class Main {
    public static void main(String[] args) {
        int[] numbers = {1, 2, 3, 4, 5, 6};
        int total = 0;
        for (int num : numbers) {
            if (num % 2 == 0) {
                total = total + num;
            }
        }
    System.out.println("Sum of evens: " + total);
}
```

Python

```python
numbers = [1, 2, 3, 4, 5, 6]
total = 0

for num in numbers:
    if num % 2 == 0:
        total = total + num

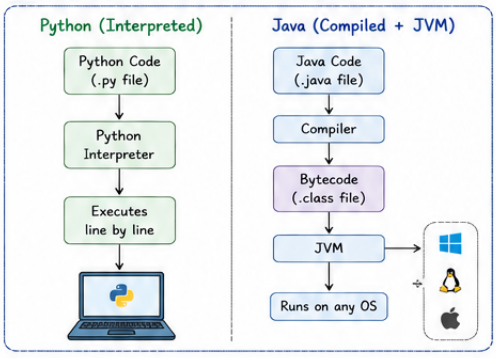
print("Sum of evens:", total)
```

Things to Notice

- ✓ Java needs a class and a main() method.
- ✓ Java uses data types (e.g., int).
- ✓ Python infers data types automatically.
- ✓ Java uses System.out.println().
- ✓ Python uses the simple print() function.
- ✓ Python code is shorter and cleaner.

Remember: Python's goal is readability. Its syntax is designed to be simple and easy to understand.

2 Python Interpreter vs Java JVM



Python

- Runs using an interpreter.
- Executes the program one line at a time.
- Very quick to write and test programs.
- Excellent for beginners and rapid development.

Java

- Source code is first compiled into Bytecode.
- The Java Virtual Machine (JVM) executes the Bytecode.
- This allows Java programs to run on different operating systems without changing the code.

Java slogan: Write Once, Run Anywhere.

3 Limitations of Python

Slower execution
Interpreted code is generally slower than compiled languages like Java and C++.

Runtime errors
Some mistakes are detected only while the program is running.

GIL in CPython
The Global Interpreter Lock (GIL) limits true multi-threading in CPU-bound tasks.

Distribution
The target computer usually needs Python installed to run the program.

Example: Runtime Error

```python
user_score = 95

if user_score > 50:
    print("You passed!")

print("Final score:", usr_score)    <- typo
```

The variable name is misspelled on line 6!

- ✓ user_score
- ✗ usr_score

What happens?

Program starts → Runs successfully until line 6 → Stops with an error → `NameError: name 'usr_score' is not defined` (Error message)

Key Point: Python executes code line by line and reports errors only when it reaches the problematic line.

Quick Summary

- Python is easy to read and write.
- Python uses an interpreter to run code line by line.
- Java uses a compiler and JVM to run programs anywhere.
- Every language has its pros and cons.

Created with the help of ChatGPT by Mandar Vaze